~~DGS~~ CGM (M&E) Pl discuss
7/9/2017

Pipeline Head Office
Maintenance Department

PLHO/MNT/MECH/05 **August 31, 2017**

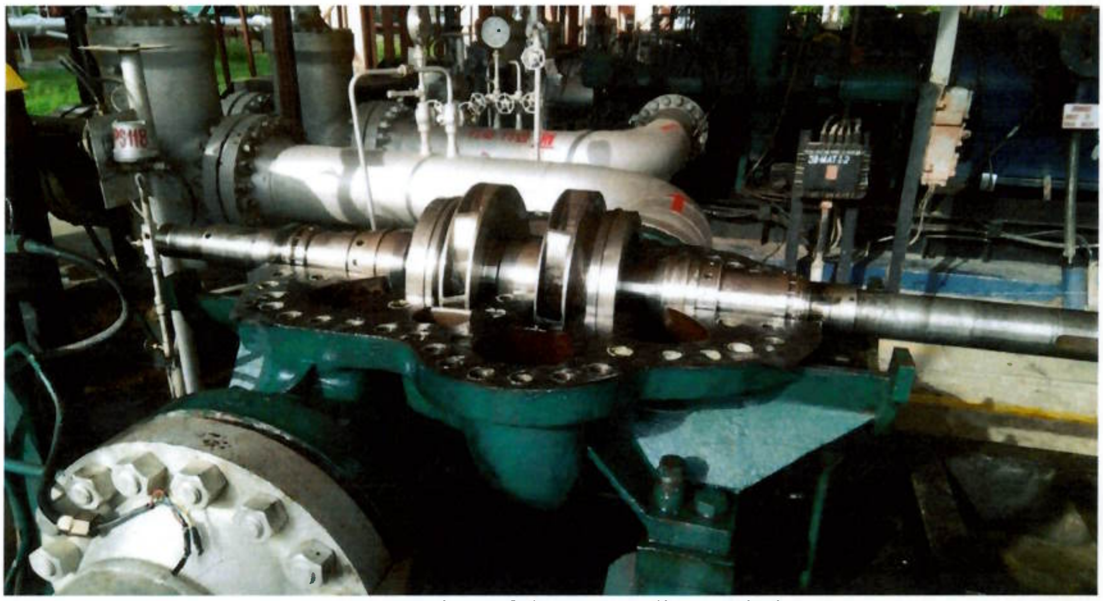
Sub: Inspection and analysis report on failure of MP-1 MJPL Mathura

A. Background

MP-1 at Mathura tripped in DE side pump bearing temperature hi-hi at 00:34:45 hrs on 09.08.2017 at CRH 160923. At the time tripping, MS was being pumped at flow rate of 556 Kl/hr. In preliminary observations, pump DE side ball bearing cage found damaged and following activities were carried out from 10.08.17 to 12.08.17.

- Inner case of failed ball bearing extracted after cutting it through grinder.
- Bearing sleeve was extracted with use of puller.
- New bearing sleeve fitted back.
- New ball bearing fitted.
- Bearing housing assembled.

After assembly of the above mentioned parts, pump shaft was not rotating freely, hence it was decided to open the pump casing for further investigation. Markings were observed in wear ring, center bush and neck ring. Pump shaft and other components sent to M/s Press tools Industries Pvt. Ltd. Ghaziabad for refurbishment work.

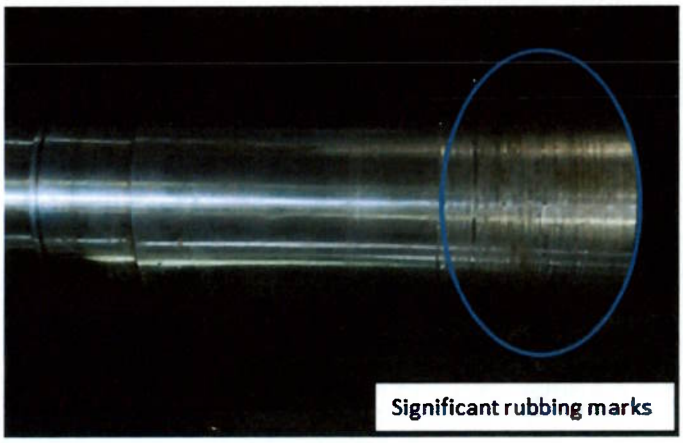


Upper casing of the pump dismantled

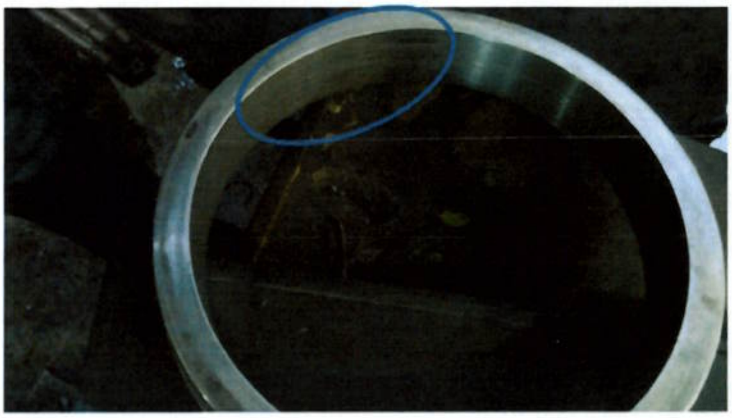
B. <u>Inspection of rotor assembly at Workshop in Ghaziabad</u>

Undersign visited the M/s Press tools Industries Pvt. Ltd. Ghaziabad on 19.08.2017 and carried out the inspection of rotor assembly. Details are placed below:-

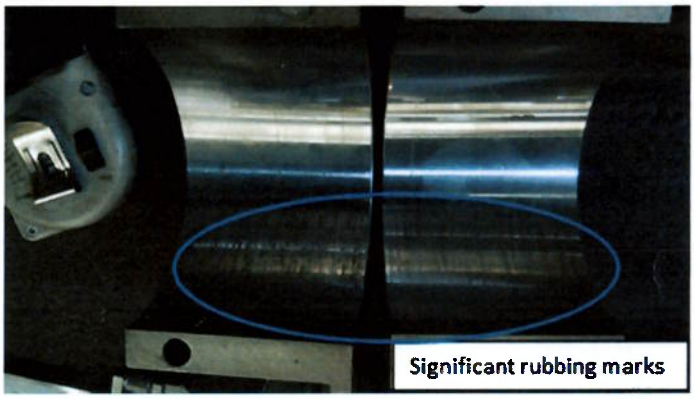
1. Rubbing marks were observed between following stationary and rotatory components :
    - Seal sleeve and throat bush

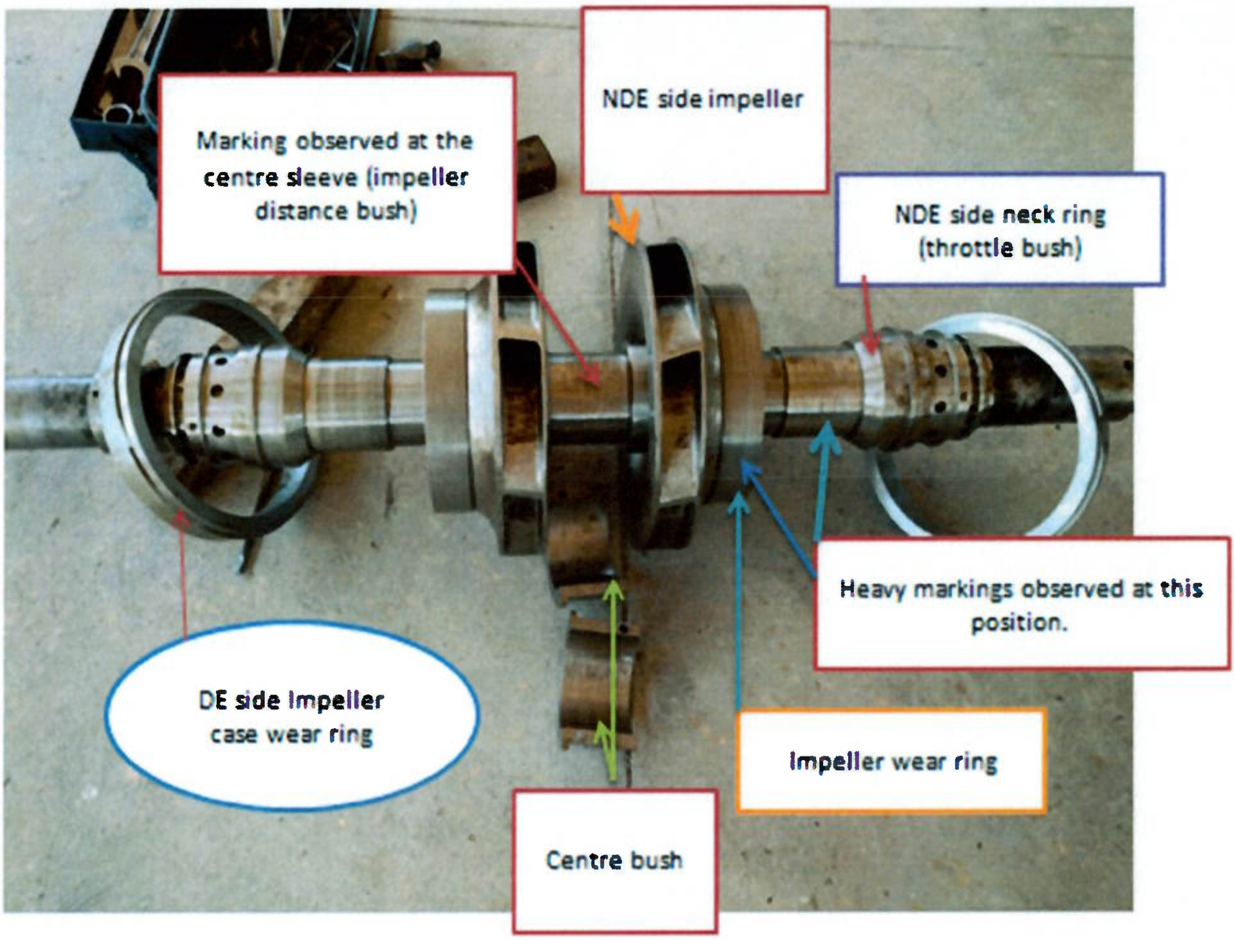


    - Impeller wear ring & case wear ring



    - Center bush and guide

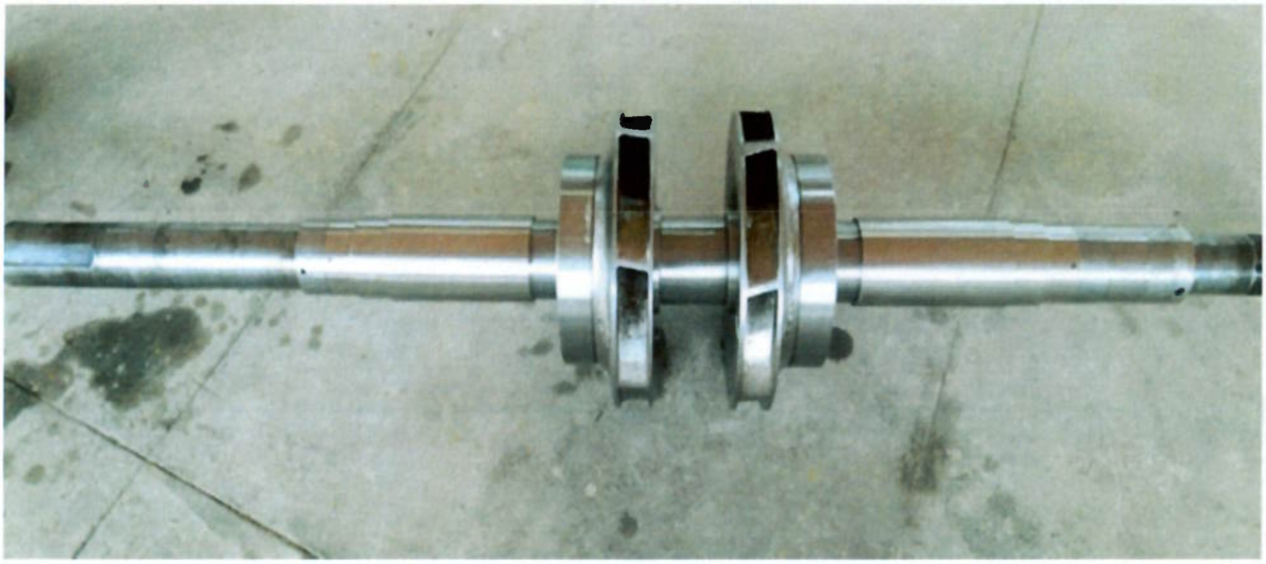




Rotor assemply of pump

2. No seizure or heat marks observed on the surface of pump shaft, stationary or rotatory components like impeller wear ring, case wear ring, throat bush, seal sleeve, center bush.
3. Shaft, case wear ring, throat bush, center bush and center bush guide can be reused. Significant rubbing marks were observed on seal sleeve so it cannot be reused.
4. Works carried out at workshop in Ghaziabad are
   - Shaft trueness checking completed and no run out or bend observed in the shaft.
   - Fabrications of one impeller wear ring and fitting of both wear rings.
   - Polishing of shaft.
   - Final grinding of pump components, case wear ring, throat bush, center bush & guide to remove the rubbing marks.
   - Clearance measurement between all the stationary & rotatory components. Clearance details are placed below.
     - ❖ Impeller wear ring & Case wear ring (0.60 mm)
     - ❖ Seal sleeve & Throat Bush (0.70 mm)
     - ❖ Center stage bush & Guide (0.50 mm)

- Assembly and Balancing of pump rotor assembly.



C. Observations on occurrence of breakdown

Visited MJPL Mathura on 25.08.2017 and reviewed the operation and maintenance practices being followed, adherence to OEM guideline in maintenance, MLPU parameters like current, bearing temperature, vibration etc before failure, Schedule & Breakdown details of MP1, Inspection of the damaged ball bearing etc. Observations are placed below:-

1. MP1 MJPL Mathura name plate details are placed below:

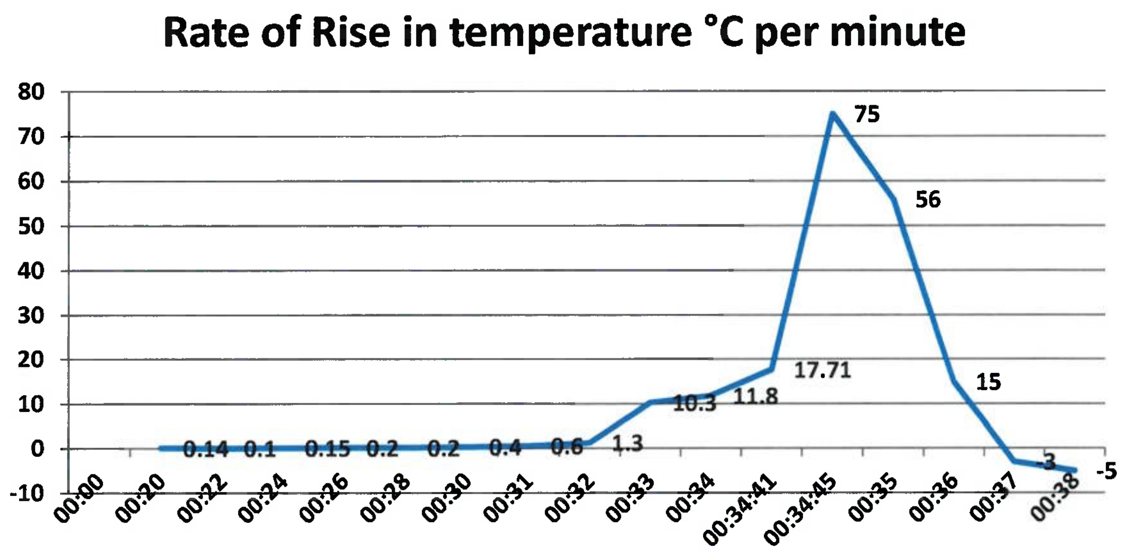
Station	Mathura	**Date -Installed & commissioned**	1982
Prime mover	Motor	**Pump Make**	BPCL
Motor Make	ACEC	**Pump Model**	DVSS 10x10x14E-2
Motor Country/state	Belgium	**Pump RPM**	2980
HZ	50	**Stages in Pump**	2
Supply Voltage-KV	6.6	**Pump No**	
Full load current	70.7	**Pump Capacity (Kl/Hr)**	568
VFD	No	**Pump Head (MCL)**	342.5

2. The last major maintenance of pump was carried out in the month of Sep'16 at CRH 157274 and Pump Bearings were replaced at that time along with other maintenance work. Ball bearing used during maintenance was of NTN make and completed 3649 Hrs before failure.

3. Last monthly maintenance was carried out on 29th July at 160751 CRhrs when lube oil (SS-46) was replaced.

4. MP – 1 was started on 08.08.2017 at 0811 Hrs and at 00:35 Hrs. on 09/08/2017 i.e. after running 16 Hrs. 24 min. MP – 1 tripped. Pump DE side bearing temperature increased from 49°C to 90°C within 3-4 minutes. Details of temperature trend is placed below:

SN	**Time**	**DE side Pump Bearing temperature °C**	**Rise in temperature °C per minute**	**Remarks**
1	00:00	44.0 ^{0}C		**Minor rise in temperature ie 4.9 °C in 30 minutes.**
2	00:20	46.8 ^{0}C		
3	00:22	47.0 ^{0}C	0.1	
4	00:24	47.3 ^{0}C	0.15	
5	00:26	47.7 ^{0}C	0.2	
6	00:28	48.1 ^{0}C	0.2	
7	00:30	48.9 ^{0}C	0.4	
8	00:31	49.5 ^{0}C	0.6	
9	00:32	50.8 ^{0}C	1.3	**Increase in Rate of temperature rise with every minutes indicates failure of ball bearing**
10	00:33	61.1 ^{0}C	**10.3**	
11	00:34	72.9 ^{0}C	**11.8**	
12	00:34:41	85^{0}C (Hi Alarm)	**17.7**	
13	00:34:45	90^{0}C (Hi – Hi Alarm. MP -1 tripped)	**75**	
14	00:35	104^{0}C	**56**	
15	00:36	119^{0}C		**Maximum temperature 119.29 at 00:35:52 hrs.**
16	00:37	116^{0}C		
17	00:38	111^{0}C		



5. Bluish heat marks were observed on the inner & outer cage surface and also found deformed. Oiler ring was found damaged. Inner race of ball bearing can only be removed from the sleeve after being cut manually.

6. Reading of the vibration is being taken regularly by station maintenance group and vibration analysis is done by the vendor deployed in Mathura Refinery. In the vibration analysis report carried out in Jan' 17 vendor has given remark "**within trend**". Last Vibration reading of the pump was taken on 07.08.2017 and DE side vibration reading is more than the normal. Details are placed below:

Location	H	V	A
Pump-NDE	7.3	5.8	4.4
Pump-DE	13.9	8.6	8.7
Motor-DE	1.8	1.4	1.6
Motor-NDE	1.1	0.7	0.9

7. DE side Pump vibration is high since last maintenance carried out in September 2016. However there was not a single instance of bearing or mechanical seal failure in this pump since September 2016. Indicative history of pump vibration as collected is given below.

Date	Flow rate (Kl/Hr)	Pump NDE			Pump DE		
		H	V	A	H	V	A
26-Aug-16	586	6.7	5.4	5.0	29.3	21.0	17.2
30-Sep-16	585	3.0	4.4	7.4	11.7	5.0	8.0
20-Jan-17	595	7.1	5.4	3.6	20.0	5.4	6.2
10-Feb-17	490	7.8	5.6	4.6	23.3	16.7	16.6
27-Mar-17	590	7.6	4.0	3.0	16.0	10.4	8.6
7-Apr-17	600	7.0	5.5	4.5	16.5	10.5	13.5
17-May-17	505	8.3	4.7	3.8	22.0	17.0	11.5
22-Jun-17	615	7.0	4.5	2.9	14.4	8.4	10.6
11-Jul-17	540	7.8	5.0	3.8	19.0	11.8	12.6

8. Balls and cage of DE side pump Ball bearing were not found hence could not be inspected for physical condition.

9. Oil was found in the bearing sump and constant oil leveler. Prior to this failure no abnormality was reported or found. However oil top up details and records were not found in station.

10. Motor current trend placed below indicates that the there is no variation in the current of the motor ie load.

11. Vibration level Hi alarm recorded in the system before bearing temperature hi alarm, placed below.

Date	Time	Point ID	Event		Level	Description	Value	Field Time
09-Aug-17	00:34:48	M1_MV111_4	CHANGE	OK	J 00	MOV_111 OPENED	NORMAL	
09-Aug-17	00:34:47	M1_MP1SEC	ALARM		U 00	MP1 SECOND STAGE ALARM	ALM	
09-Aug-17	00:34:45	M1_M1PBTHA	ALARM		U 00	MP1 PMP BRG TEMP HIHI ALM BIT	HIHI	
09-Aug-17	00:34:45	M1_MMP1_29	CHANGE	OK	J 00	MP1 INITIAL CONDITIONOK BIT	NORMAL	
09-Aug-17	00:34:45	M1_MMP1_30	CHANGE	OK	J 00	MP1 SET AVAILABLE	NORMAL	
09-Aug-17	00:34:45	M1_MMP1_33	CHANGE	OK	J 00	MP1 SEQUENCE ON BIT	NORMAL	
09-Aug-17	00:34:45	M1_MMP1_22	ALARM		U 00	MP_1_4 PUMP BRG TEMP HH (DE)	HIHI	
09-Aug-17	00:34:45	M1_MP_1_23	ALARM	OK	U 00	MP_1 VIB. LVL. HI	NORMAL	
09-Aug-17	00:34:45	SINGPMPRUN	ALARM		J 00	MJPL-SINGLE PUMP RUNNING	RUN	
09-Aug-17	00:34:45	DOUBLEPMPRUN	ALARM	OK	J 00	MJPL DOUBLE PUMP RUNNING	NOTRUN	
09-Aug-17	00:34:41	M1_MP1FST	ALARM		U 00	MP1 FIRST STAGE ALARM	HIGH	
09-Aug-17	00:34:41	M1_MMP122A	ALARM		U 00	MP_1_4 PUMP BRG TEMP HI (DE)	HI	
09-Aug-17	00:34:40	M1_MP_1_23	ALARM		U 00	MP_1 VIB. LVL. HI	HIGH	
09-Aug-17	00:34:37	M1_MP1FST	ALARM	OK	U 00	MP1 FIRST STAGE ALARM	NORMAL	
09-Aug-17	00:34:35	M1_MP_1_23	ALARM	OK	U 00	MP_1 VIB. LVL. HI	NORMAL	
09-Aug-17	00:34:31	M1_MP1FST	ALARM		U 00	MP1 FIRST STAGE ALARM	HIGH	
09-Aug-17	00:34:30	M1_MP_1_23	ALARM		U 00	MP_1 VIB. LVL. HI	HIGH	
08-Aug-17	23:32:53	M1_DC_MAIN	OP FAIL		J 00	DECREMENT IN MANUAL MODE	DCR	

D. Conclusion

- From the above details it is inferred that the breakdown of MP1 MJPL Mathura occurred due to failure of pump DE side ball bearing and bearing failure was due to starvation of lubricant oil in bearing.

- DE side pump bearing data indicates rise in bearing temperature started around 00:00 hrs and first stage alarm at 00:35:41 hrs. The temperature rise pattern indicates lack of lubrication.

- After failure of ball bearing the rubbing between stationary and rotatory components of rotor assembly have taken place.

- Further, previous data indicates persistence of high, more than 14mm/sec and vibration level in DE side of pump which may also have contributed to bearing cage failure.

- An alarm based on rate of rise in bearing temperature will be helpful in averting such incidence.

(Amit Kumar)
SMNM
31/8/2017

GM(Maintenance) :

- There is substantial indication to infer that bearing failure was caused by lack of lubricating oil aided by high vibration.
- The recording about lube oil top up in pumps should be a regular practice to enable monitoring of lube oil quality, quantity and top up frequency.

- The lube oil top up record to be checked by shift officer in regular interval to check the lube oil is being consumed in regular pattern. In case oil level remains constant for some time, the oil flow must be checked for any suspected blockage.
- The presence of high vibration especially in the DE side over prolonged time needs to be analyzed along with supports in the suction & discharge arm.
- After such failure we should preserve lubricating medium and damaged components for testing and analysis to establish the initiation of failure.

1-9-2017

S.P.Pandey

CGM (M&I): Region has been advised to open and maintain lube oil register at all station and check the suction/discharge arm supports for consistent high vibration. Upon feed back further study will be made.

1/09/2017

ED (O):

May kindly like to brief DPL

Dr. Rakesh Mishra (DGM, DPL Secretariat)

5/9

Apprised D(PL)

06/9/17